

```
# Setup the network addresses you are protecting
#
# Note to Debian users: this value is overridden when starting
# up the Snort daemon through the init.d script by the
# value of DEBIAN_SNORT_HOME_NET s defined in the
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET any
```

Figure 5: Conf variable

```
#####
# Step #6: Configure output plugins
# For more information, see Snort Manual, Configuring Snort - Output Modules
#####

# unified2
# Recommended for most installs
# output unified2: filename merged.log, limit 128, nostamp, mpls_event_types, vlan_event_types
# output unified2: filename snort.log, limit 128, nostamp, mpls_event_types, vlan_event_types

# Additional configuration for specific types of installs
# output alert_unified2: filename snort.alert, limit 128, nostamp
# output log_unified2: filename snort.log, limit 128, nostamp

# syslog
# output alert_syslog: LOG_AUTH LOG_ALERT

# pcap
# output log_tcpdump: tcpdump.log

# metadata reference data. do not modify these lines
include classification.config
include reference.config
```

Figure 6: Conf output

```
#####
# Step #7: Customize your rule set
# For more information, see Snort Manual, Writing Snort Rules
#
# NOTE: All categories are enabled in this conf file
#####

# Note to Debian users: The rules preinstalled in the system
# can be *very* out of date. For more information please read
# the /usr/share/doc/snort-rules-default/README.Debian file

#
# If you install the official VRT Sourcefire rules please review this
# configuration file and re-enable (remove the comment in the first line) those
# rules files that are available in your system (in the /etc/snort/rules
# directory)

# site specific rules
include $RULE_PATH/local.rules
```

Figure 7: Conf rules

file, restart the application and we have a new working configuration.

Before going any further, let's take a brief look into the syntax of Snort rules.

- Snort rules must be contained in a single line or we can use the multi-line character \. For example:

```
log tcp !x.x.x/xx OR
log tcp !x.x.x/xx any -> xxx \
(msg: "some command")
```

- All rules should contain a rule header (which identifies the actions) and rule options (which identify the rule's alert messages).
- The rules must describe situations like a violation of the security policy of the company, or correctly detect the exploitable vulnerabilities.
- There are three kinds of rules in Snort:
 - **Alert rules:** This generates

alerts using the alert method.

- **Log rules:** Upon generation of any alert, it logs that specific alert.
- **Pass rules:** Ignores the packet if deemed malicious and drops it.

Now we can move on to the configuration file, which can be opened using the following command:

```
kali > mousepad /etc/snort/snort.conf
```

In the Snort configuration file, we may see nine sections, as shown in Figure 4.

For the most basic configuration, we will address only Sections 1, 6 and 7 as seen in Figure 5, 6, 7.

Setting variables

In the screenshot in Figure 5, we can see the highlighted line 'ipvar HOME_NET'. This variable denotes the network is protected. 'HOME_NET' is the variable name to which the IP address is assigned. This can be a single IP address, a list of IP addresses, or a subnet in CIDR notation, or even can be left as any.

Checking the output

Using *Ctrl+F*, we then move on to output plugins (Figure 6). By default, Snort sends the output in log format. But if we want, we can comment out that line (*unified2*) and uncomment the bottom line (*log_tcpdump*), enabling the output in tcpdump format, which is saved in the */var/log/snort* directory.

Disable rules

Depending on your enterprise, we may need to change the rules that Snort relies upon, and customise them in Section 7, as shown in Figure 7.

To not let Snort use a given set, simply comment out the *include* part. After making any change, simply